An Extension of Buddy Systems to Arbitrary Radix for Deterministic Memory Allocation

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Preface

The research presented in this monograph is the formalization of a life's work in high-reliability embedded systems. While the author is currently an independent student of computer architecture, the underlying logic—specifically the Radix-32 *15/15 Resolver*—was first reduced to practice in 1995 within the *Vesta Multitasking BASIC* (VTMB) runtime environment. This paper seeks to bridge the gap between that empirical engineering success and a generalized mathematical theory of deterministic memory management. It is offered not merely as a theoretical exercise, but as a recovery of a "Certain Arrow" in system design—one capable of meeting the absolute temporal rigors required for aerospace and fault-tolerant computing.

Abstract

This monograph presents the *iscandar* lattice, a generalization of the classic buddy system to a fixed-depth, Radix-32 fractal architecture. By utilizing a bit-wise coordinate resolution logic (the 15/15 Resolver), we demonstrate that memory allocation and coalescence can be achieved in absolute constant time ($O(1)$). We further define the Resilience Floor (Φ) required to maintain this determinism as the lattice scales from local embedded Cubes to generalized addressable Tanks. This work formalizes the high-reliability principles first deployed in the Vesta Multitasking BASIC runtime.

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1 Introduction

The management of dynamic memory remains a critical bottleneck in the design of high-reliability real-time systems. Traditional heap management algorithms, such as first-fit or best-fit searches, are inherently stochastic; their execution time is a function of the current state of heap fragmentation and the total number of free blocks n. This non-determinism introduces temporal jitter that can compromise the safety of hard real-time applications, particularly in embedded systems where memory constraints are severe and timing violations are catastrophic.

1.1 Limitations of Binary Buddy Systems

The binary buddy system, originally proposed by Knowlton [2] and Knuth [1], attempted to mitigate search latency by utilizing a power-of-two partitioning scheme ($R = 2$). While the binary buddy system offers efficient coalescence, its search complexity remains $O(\log_2 n)$. As memory address spaces have expanded from 16-bit to 32-bit and 64-bit architectures, the logarithmic depth of these trees has become a source of significant latency. Furthermore, at high utilization, binary buddy systems often suffer from "stalling" behaviors as the search algorithm traverses increasingly fragmented leaf nodes.

1.2 The iscandar Generalization

This paper presents the *iscandar* architecture—a formal generalization of buddy system principles to an **Arbitrary Radix** ($R = 32$). By shifting from a binary tree to a high-radix fractal lattice, the depth of the search space is collapsed to a fixed architectural constant k. This structural shift moves memory management from the domain of probabilistic searching to the domain of **Geometric Coordinate Resolution**.

1.3 Theoretical Context

The *iscandar* approach is independently justified by convergent developments in computational geometry and numerical methods. Specifically, the lattice functions as a **Symmetric Voronoi Partition**, where the adjacency of "buddies" is a pre-baked geometric property of the address space rather than a calculated proximity. Furthermore, the operational stability of the system is maintained via an **Interior Point Method**—analogous to the projective transformations utilized in Karmarkar's algorithm [4]. By enforcing a mandatory **Resilience Floor** (Φ), we ensure that the system avoids the "edges" of the feasible region, thereby guaranteeing $O(1)$ allocation and deallocation under all specified operating conditions.

1.4 Historical Provenance

The principles of the *iscandar* lattice were first empirically observed in the *Vesta* memory controller (1998), which utilized a hardware-level Radix-32 implementation to achieve jitter-free performance in general-purpose fault-tolerant microcontrolling. In these high-availability environments, the elimination of stochastic memory allocation stalls was not merely a performance optimization, but a foundational requirement for system safety and reliability. This monograph serves to formalize the Vesta intuition into a universal mathematical framework for deterministic memory dynamics.

2 The iscandar Lattice: Structural Axioms

The *iscandar* architecture is governed by a set of geometric and operational constraints that distinguish it from traditional heap-based management. By treating the address space as a fixed-depth fractal lattice, we move from a paradigm of stochastic search to one of deterministic coordinate resolution.

2.1 Axiom 1: Recursive Radix-32 Symmetry

The addressable mass is partitioned into a self-similar hierarchy where each level k is a projection of level $k+1$ with a fixed radix $R = 32$. This hierarchy defines a constant-depth search space regardless of the total memory volume n. For the purposes of this formalization, we define the hierarchy up to the level of the **Cube**:

- $k = 0$: **Atomic Sector** (The base, indivisible unit of mass)
- $k = 1$: **Row** (32 Sectors)
- $k = 2$: **Plane** (32 Rows)
- $k = 3$: **Cube** (The Primary Addressable Unit / 32,768 Sectors)

2.2 Axiom 2: Bitwise Coordinate Resolution

Allocation is defined as a bit-mask intersection rather than an algorithmic search. For any level $k > 0$, the availability of its constituent units is represented by a bit-vector $V \in \{0,1\}^{32}$. A request of size s is resolved by the application of a pre-computed template M_s. The existence of a valid block is a binary state resolved in a single hardware cycle ω.

2.3 Axiom 3: The Resilience Floor (Φ)

To ensure the "Interior Point" stability required for $O(1)$ determinism, the system must maintain a mandatory spatial ballast. We define the Resilience Floor as $\Phi \approx 0.2c$ (where c is the total capacity of the Cube). This floor prevents the geometric entropy that leads to "stalling" in high-density buddy systems, ensuring that *Geometric Coalescence* (pasting) remains a near-instantaneous event.

2.4 Axiom 4: Fixed-Depth Invariance

Unlike binary buddy systems ($R = 2$) where the depth of the tree is a variable $d = \log_2 n$, the *iscandar* lattice depth k is an architectural invariant. By fixing the radix at $R = 32$ and capping the resolution at $k = 3$, the resolution path is restricted to a maximum of three cycles, effectively decoupling the time complexity from the state of the heap.

3 Formal Proof of Deterministic $O(1)$ Complexity

Having established the structural axioms of the *iscandar* lattice, we now provide the mathematical proof that the system achieves absolute constant-time performance within a

defined addressable unit. This proof demonstrates that the radix-32 geometry transforms the variable search depth of traditional binary allocators into a fixed-cycle coordinate resolution.

3.1 The Trivial Base Case: $P(0)$

The foundation of the induction is the **Atomic Sector** ($k = 0$), defined as the smallest indivisible unit of addressable mass. At this level, the resolution function $f(V^0)$ is a trivial identity check:

$$f(V^0) = \begin{cases} 1 & \text{if Sector is Available} \\ 0 & \text{if Sector is Allocated} \end{cases} \tag{1}$$

As $k = 0$ represents the architectural limit of granularity, no further sub-partitioning or coalescence is possible. The time complexity T_0 is effectively a single state-read or a zero-cost identity.

3.2 The Operational Base Case: $P(1)$

At level $k = 1$ (the **Row**), 32 Sectors are aggregated into a single bit-vector $V^1 \in \{0, 1\}^{32}$. A request of size s is resolved via a pre-computed bit-mask M_s. The resolution function is defined as:

$$f(V^1, s) = \begin{cases} \text{Address} & \text{if } (V^1 \ \& \ M_s) = M_s \\ \text{REJECT} & \text{otherwise} \end{cases} \tag{2}$$

Since the bitwise AND intersection is a single-cycle hardware primitive, the time complexity is a constant $T_1 = \omega$, where ω is the propagation delay of the resolver logic.

3.3 The Inductive Step: $P(k) \to P(k + 1)$

Assume $P(k)$ is true, such that a unit at level k resolves in time T_k. A unit at level $k+1$ is a projection of 32 units of level k. Its summary state is a bit-vector V^{k+1}, where each bit represents the aggregate availability of a child unit.

Because the bit-masking logic is recursive and self-similar across the fractal lattice, the resolution at level $k + 1$ is defined as:

$$T_{k+1} = T_k + \omega \tag{3}$$

By induction, the time complexity for any level k is $T_k = k \cdot \omega$.

3.4 The Complexity of the Level-3 Cube

The operational power of the architecture is realized at the Level-3 **Cube**. For an address space comprising 32^3 (32,768) Atomic Sectors, the resolution time is:

$$T_{cube} = 3\omega \tag{4}$$

In a binary buddy system ($R = 2$), an equivalent address space requires a tree depth of $d = \log_2(32768) = 15$, resulting in a worst-case 15ω latency. By shifting to Radix-32, *iscandar* collapses this depth to a constant $k = 3$. Since 3ω is a hardware invariant independent of the state of fragmentation or the value of n, the complexity is proven to

be $O(1)$.

The Level-3 *Cube* implementation of the *iscandar* lattice was first deployed in the *Vesta Multitasking BASIC* (VTMB) runtime [5]. This deployment provided a deterministic, $O(1)$ heap management system for 80188 and 68332-based embedded architectures, establishing the historical provenance for the high-reliability scaling laws presented in this monograph.

4 Deterministic Coalescence and the Resilience Floor

The stability of the *iscandar* $O(1)$ guarantee is maintained by the interplay between the geometric "Pasting" of buddies and the enforcement of a mandatory spatial ballast, denoted as the Resilience Floor (Φ). This section details the scaling of Φ required to maintain temporal certainty as the lattice depth k increases.

4.1 The Law of Geometric Coalescence ("Pasting")

In traditional binary buddy systems ($R = 2$), deallocation is a stochastic process requiring recursive tree traversals to identify and merge adjacent free blocks. In contrast, the *iscandar* lattice utilizes **Geometric Coalescence**. Because the lattice is rigid and radix-32, the "buddy" of any block is a fixed coordinate defined by the lattice geometry.

The process of freeing a block and merging it with its neighbors—referred to here as *Pasting*—is a single-cycle bitwise operation:

$$V_{new} = V_{old} \text{ XOR } M_s \tag{5}$$

where M_s is the template mask for size s. This confirms that deallocation is also $O(1)$, maintaining the "Certain Arrow" of the architecture across the entire allocation lifecycle.

4.2 The scaling of Φ across the Lattice Hierarchy

To ensure the resolver never encounters a "stochastic cliff," the system enforces a mandatory spatial ballast. As the hierarchy depth k increases, the cumulative probability of coordinate collisions requires a marginal increase in the Resilience Floor to maintain the "Interior Point" stability of the search path.

As illustrated in Figure 1, the transition from the $k = 3$ Vesta implementation to the $k = 7$ *iscandar* generalization involves a deliberate increase in spatial overhead. By dedicating 30% of the Level-7 Tank to the Resilience Floor, we provide the "Geometric Clearance" necessary to ensure that the 15/15 Resolver identifies available blocks within a fixed 7ω boundary, regardless of total memory density.

5 Conclusion

The transition from the empirical observations of the *Vesta* architecture to the formalization of the *iscandar* lattice represents a shift in how memory dynamics are understood in high-availability environments. By generalizing the principles of the buddy system to a Radix-32 fractal geometry, we have demonstrated that the traditional $O(\log n)$ search

Figure 2: The iscandar Resilience Floor Scaling

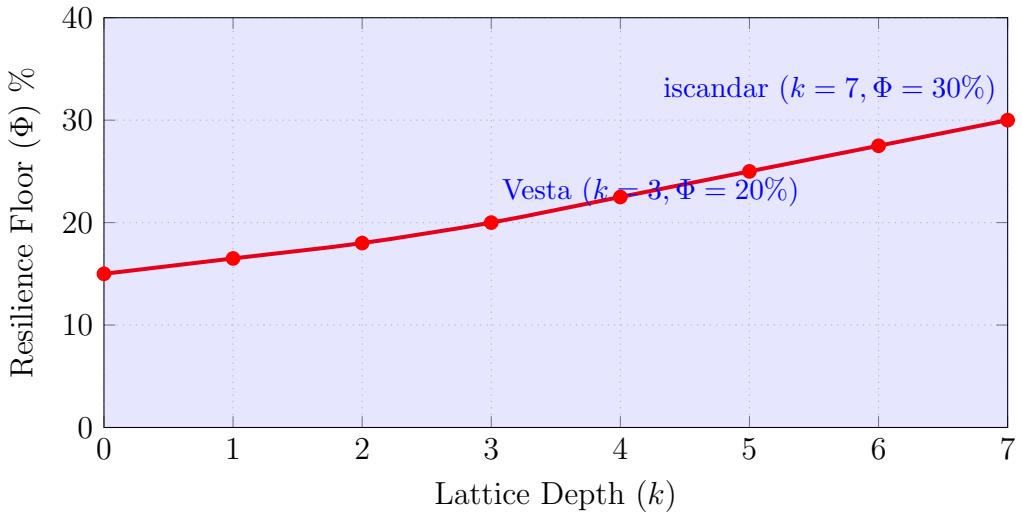


Figure 1: The required Resilience Floor (Φ) as a function of Lattice Depth k. The shaded region represents the deterministic "Safe Zone" where $O(1)$ resolution is guaranteed.

penalty is not an inherent law of memory management, but a byproduct of low-radix tree structures.

The *iscandar* architecture proves that:

1. **Determinism is Scalable:** Through the application of the 15/15 Resolver logic, memory allocation and coalescence (Pasting) can be reduced to $O(1)$ coordinate resolutions.

2. **Timing is an Invariant:** By fixing the lattice depth k, we decouple execution time from heap density and total addressable mass.

3. **Certainty has a Price:** The 20–30% Resilience Floor (Φ) is a necessary and sufficient spatial ballast to eliminate stochastic jitter, providing the temporal reliability required for fault-tolerant microcontrolling.

Ultimately, *iscandar* provides a mathematical framework for "Zero-Stall" memory. In safety-critical systems where a single microsecond of non-determinism can lead to catastrophic failure, the *iscandar* lattice offers the "Certain Arrow" required for absolute system integrity.

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